

Course: B. Tech. Branch: Computer Engineering / CSE Semester: VII

Subject Code & Name: Cryptography & Network Security (BTCOE704A)

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain classical cryptosystem and its types.	Remember	6
B) What do you mean by cryptosystem? What is its significance?	Understand	6
C) Differentiate between the symmetric key cryptography and Asymmetric key cryptography.	Remember	6
Q.2 Solve Any Two of the following.		12
A) List of Modes of operations. Explain briefly any two modes of operations.	Remember	6
B) Explain about cryptanalytic attack in detail.	Understand	
C) What is the purpose of S-boxes in DES? Explain the avalanche effect.	Remember	6
Q. 3 Solve Any Two of the following.		12
A) Differentiate between the stream cipher and Block cipher.	Remember	6
B) What do you mean by hash function? What are requirements of the hash function? Enumerate the properties of hash function.	Understand	6
C) Explain in detail Message authentication code and its requirements.	Remember	6
Q.4 Solve Any Two of the following.		12
A) Explain Diffie-Hellman key exchange algorithm in detail.	Remember	6
B) Explain ElGamal cryptosystem with an example.	Remember	6
C) Elaborate the steps of key generation using the RSA algorithm. In RSA system the public key (E, N) of user A is defined as (7,187). Calculate $\Phi(N)$ and private key 'D'. What is the cipher text for M=10 using the public key.	Analyze	6

Q. 5	Solve Any Two of the following.		12
A)	Explain digital signature standard with necessary diagram in detail.	Remember	6
B)	Explain the SSL (Secure socket layer) architecture in detail.	Remember	6
C)	How does PGP achieve confidentiality and authentication in emails?	Understand	6

***** End *****